

Target and stitched multiple-shooting estimate

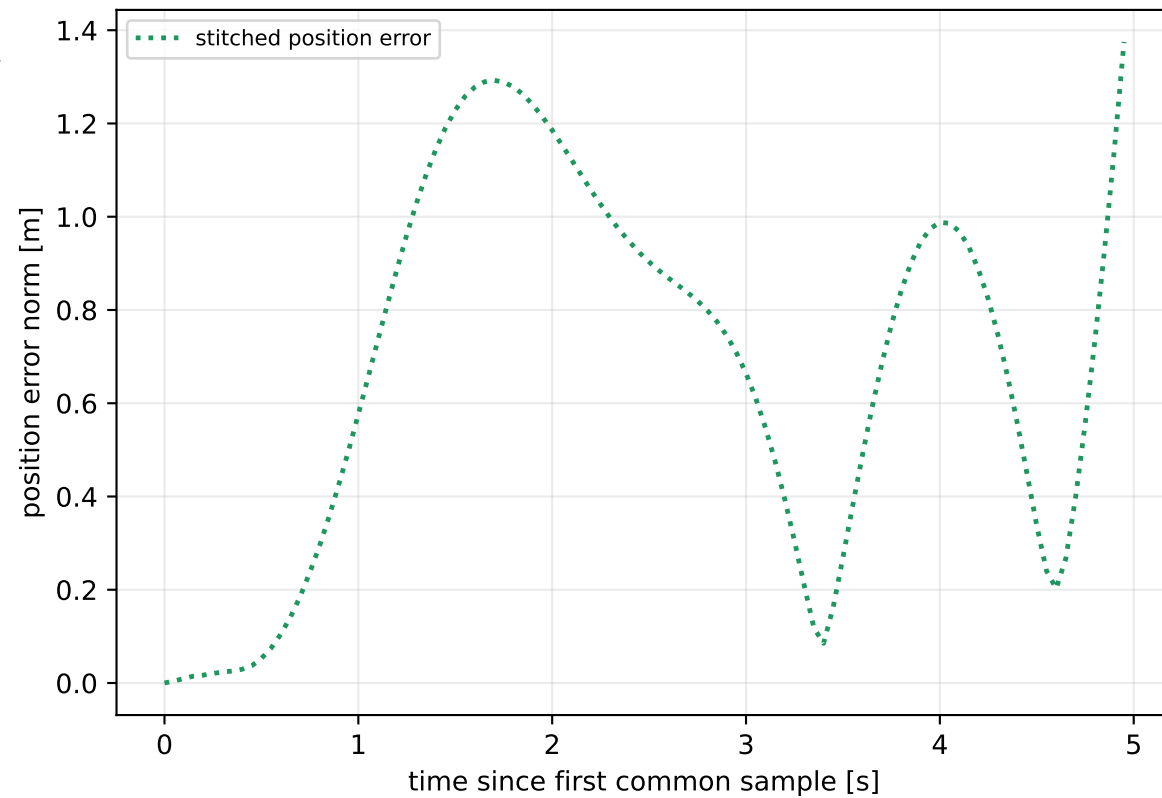
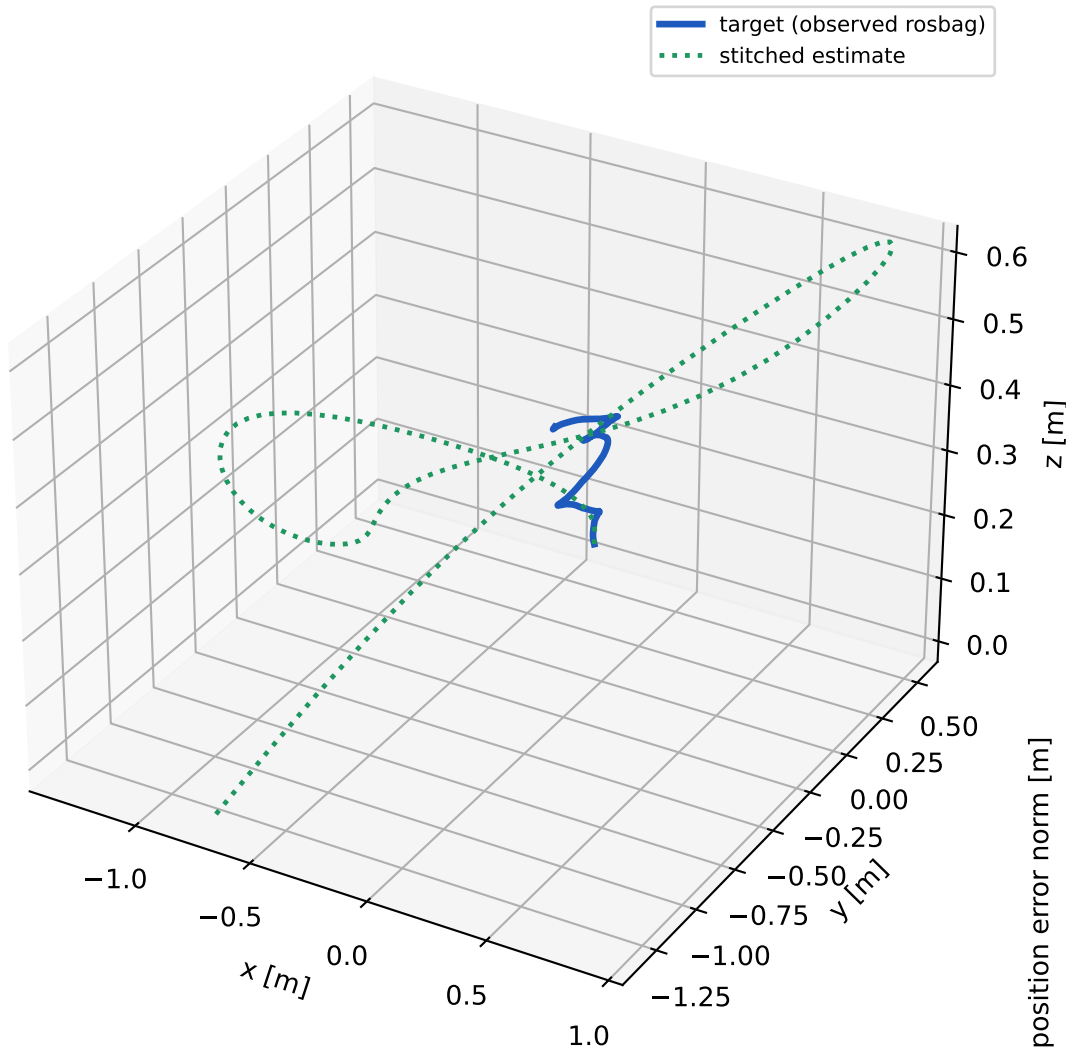
actual common support: 19.009–23.959 s
selected command delay: 0.02 s

metric	stitched
position RMSE [m]	0.78836
orientation RMSE [deg]	28.441
velocity RMSE [m/s]	1.678
angular velocity RMSE [rad/s]	1.0611
specific force RMSE [m/s ²]	1.4974

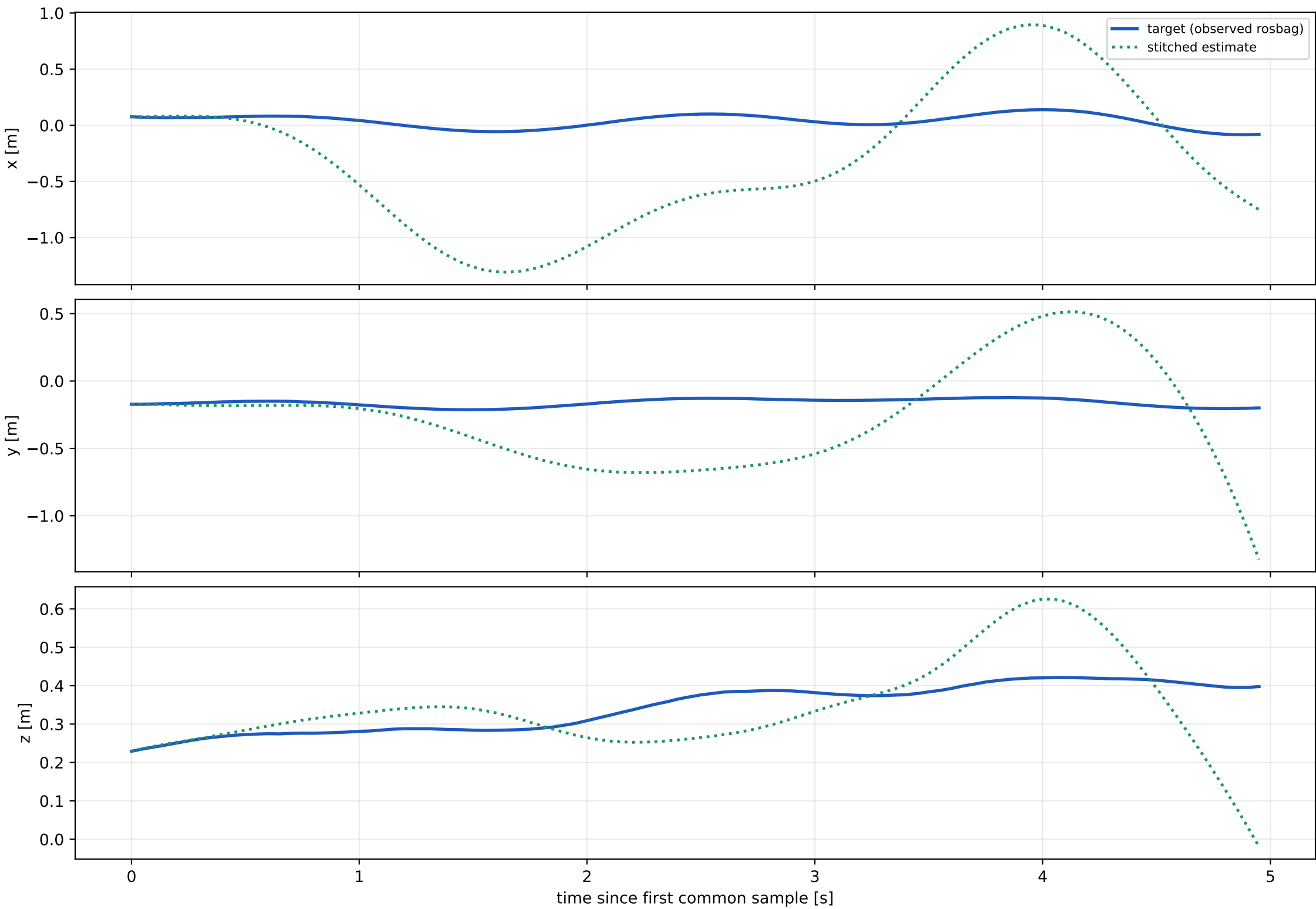
continuity max / tolerance:
0.0004449 / 0.001 (converged)

Velocity and IMU metrics are diagnostic only;
the fit uses sensor position and orientation.

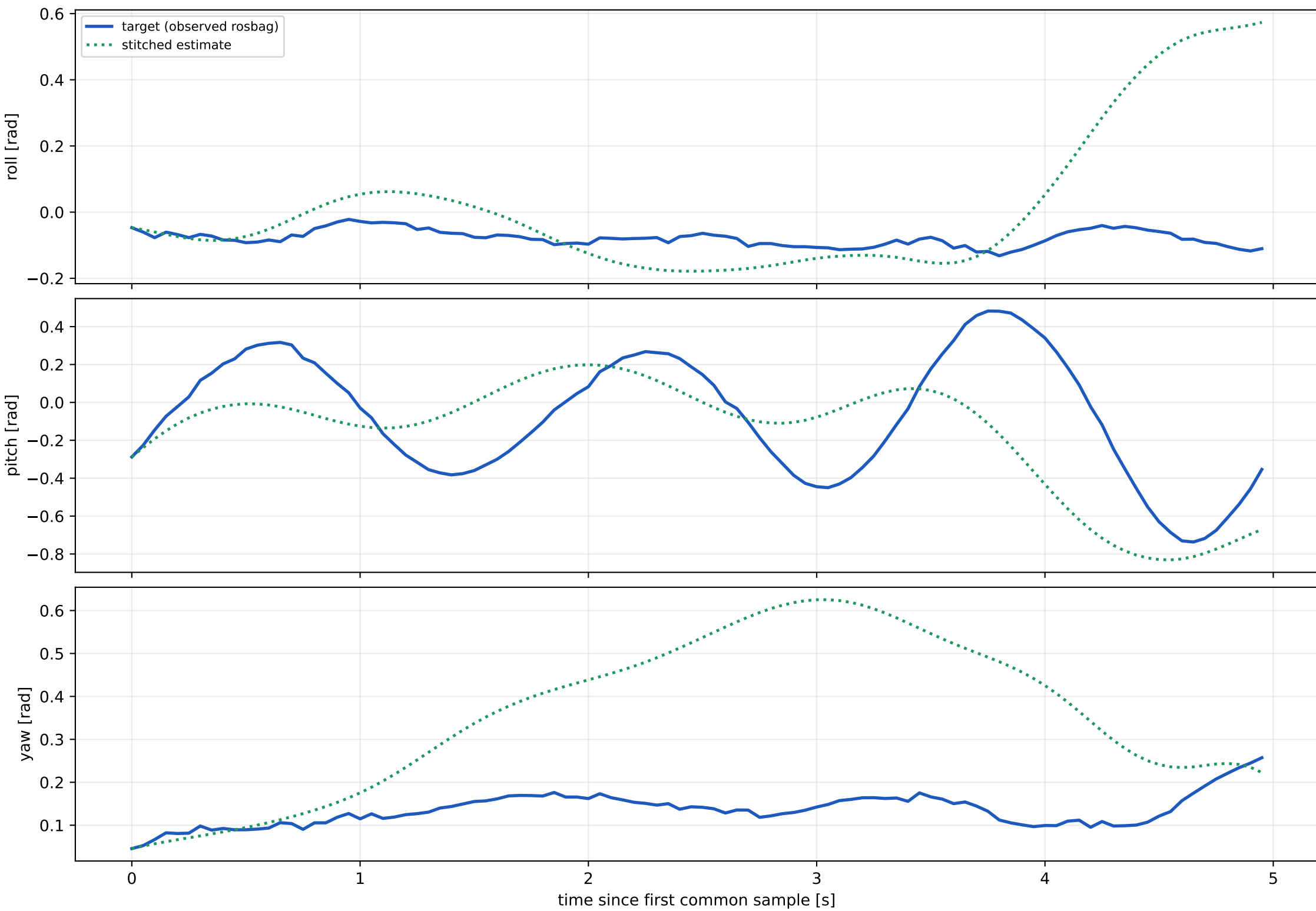
Stitched trajectory



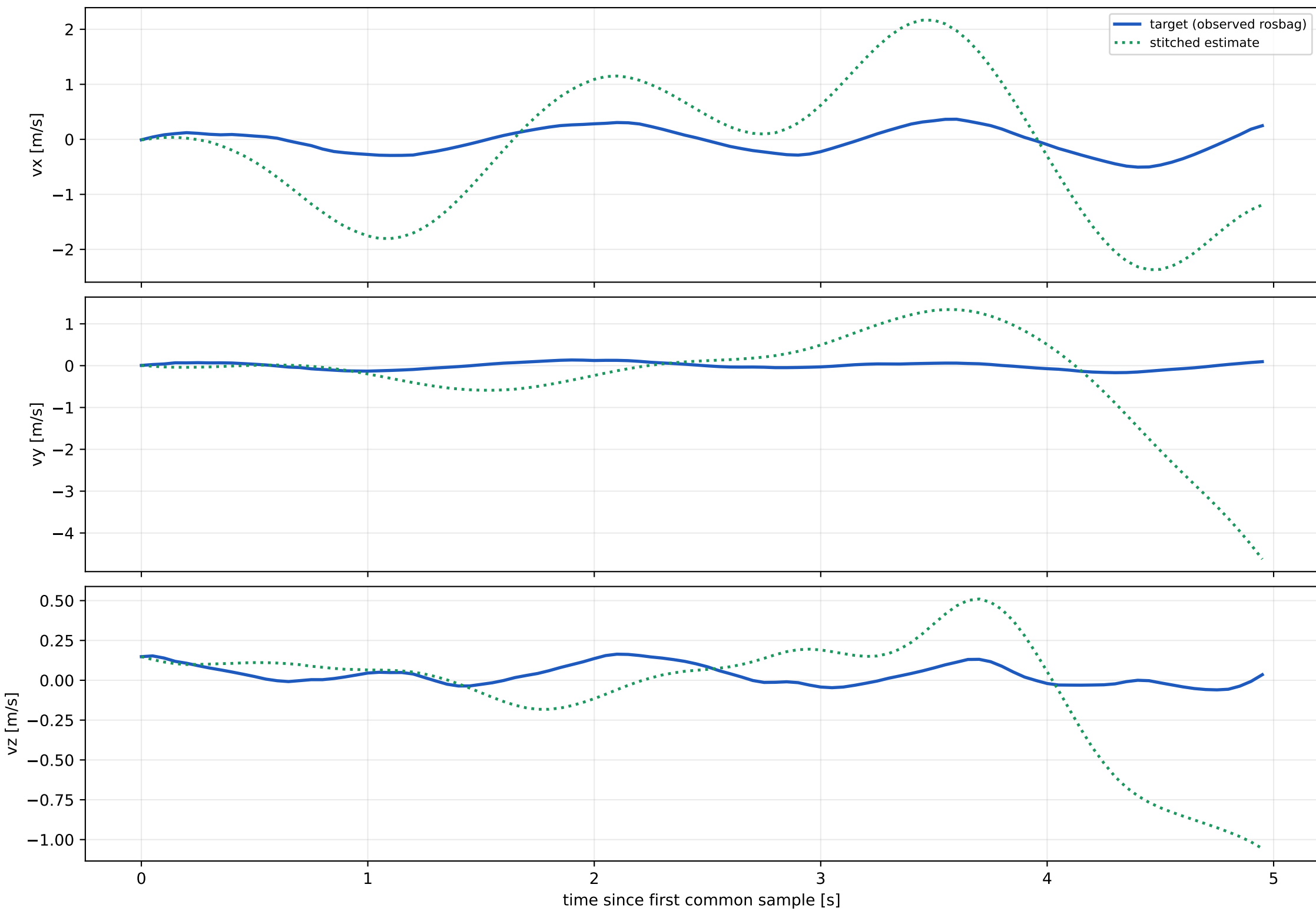
Sensor position



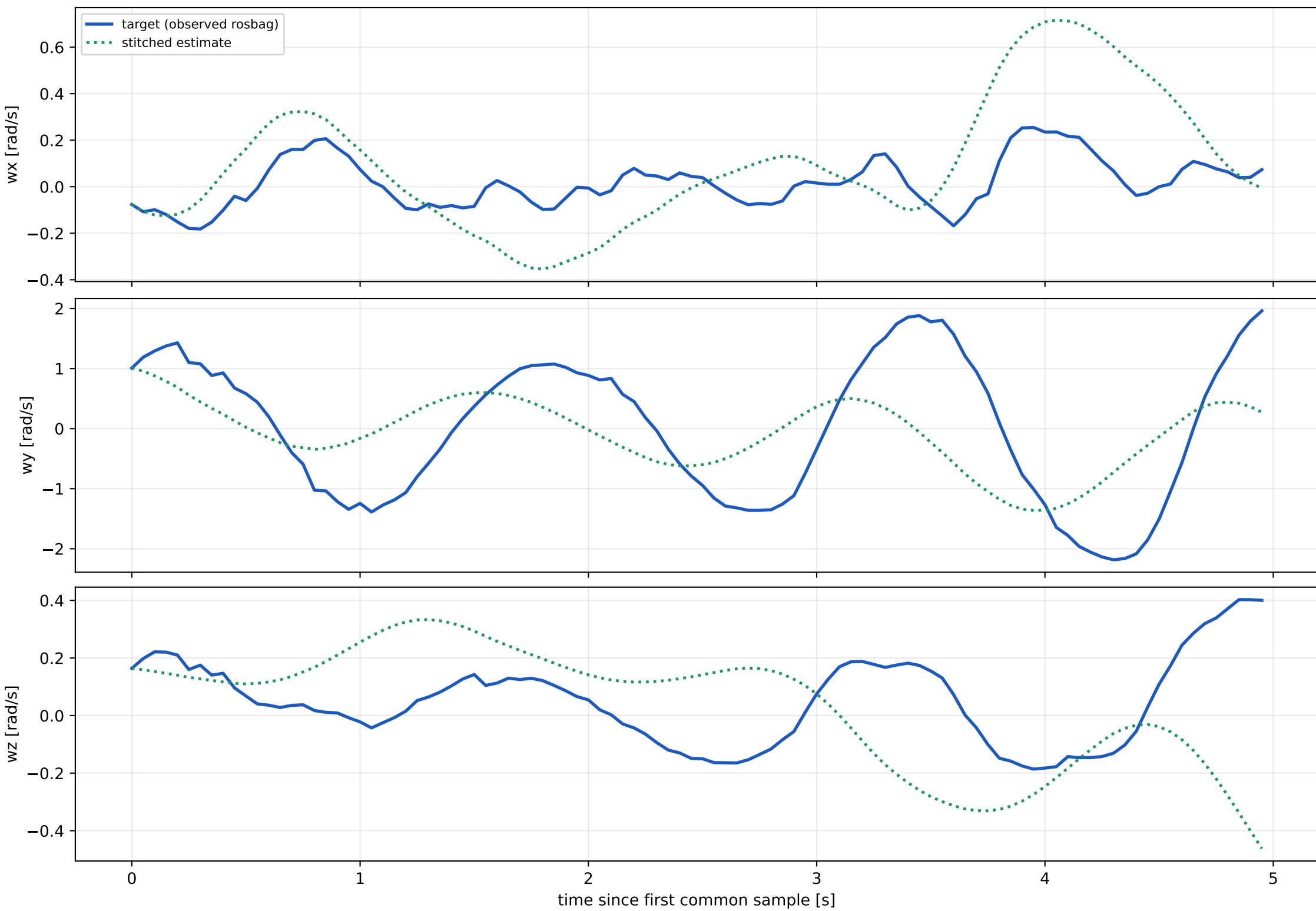
Sensor orientation (display only; fit uses SO(3) log)



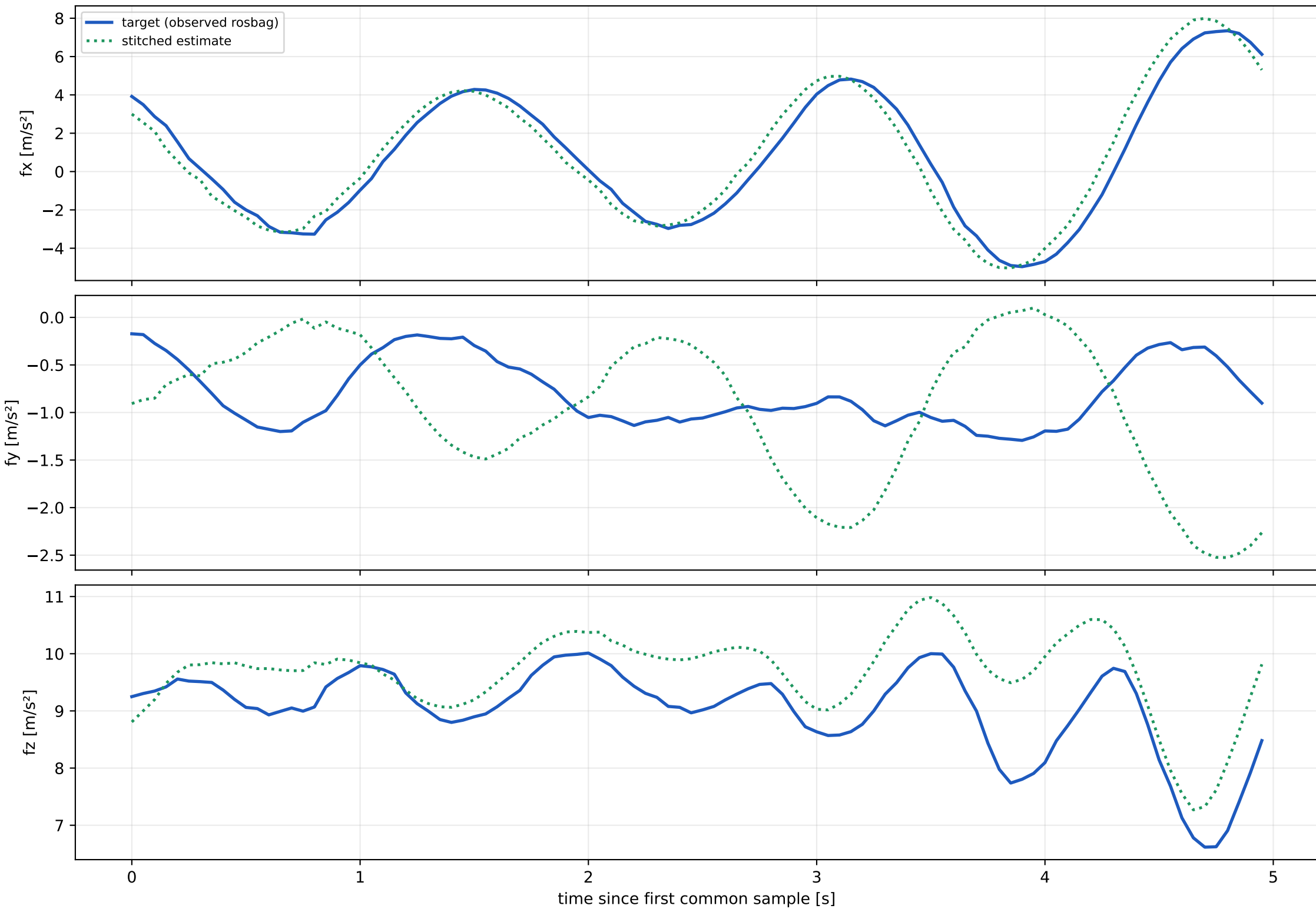
World-frame sensor velocity (diagnostic; excluded from fit)



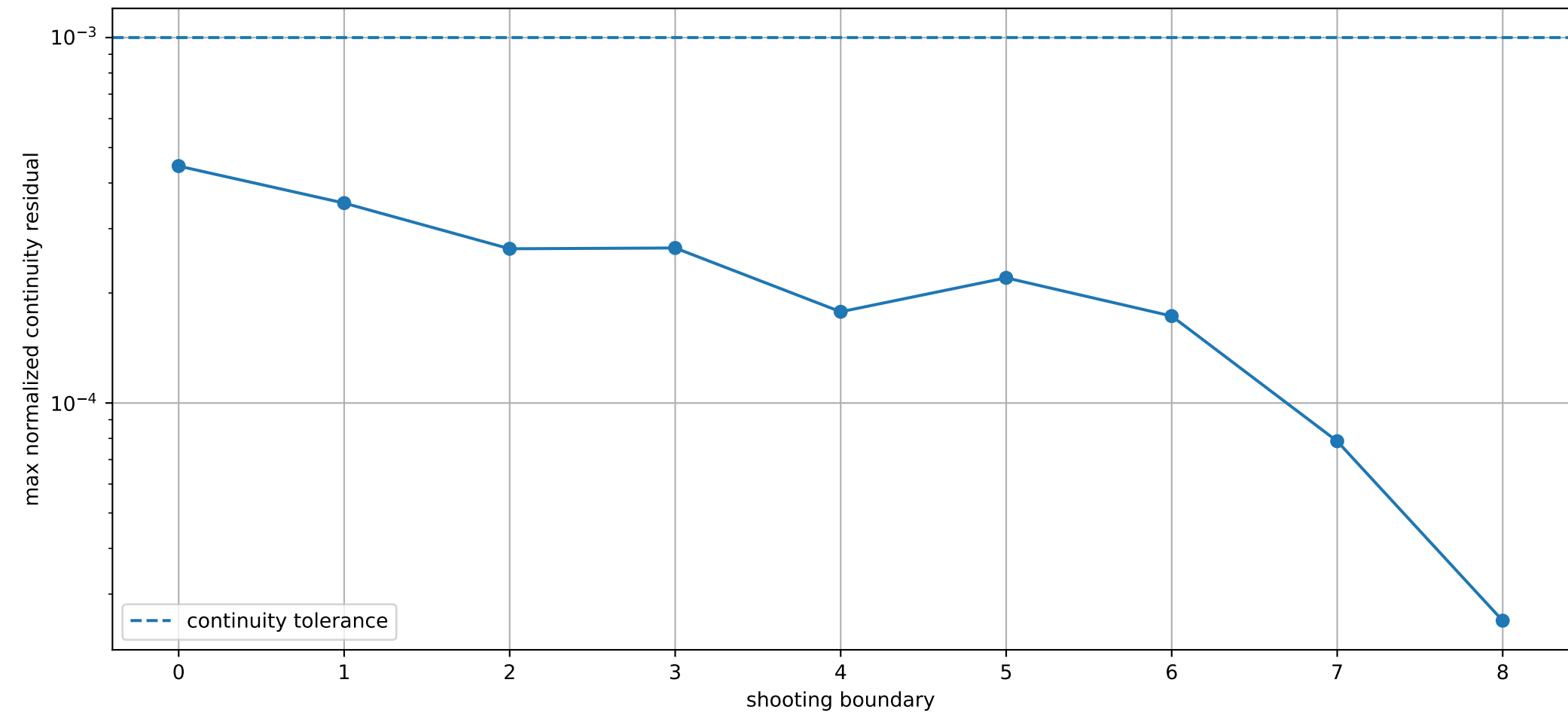
IMU angular velocity (diagnostic; excluded from fit)



IMU specific force (diagnostic; excluded from fit)



Shooting continuity diagnostic



Final estimated parameters

Selected multiple-shooting estimate

Decoded physical parameters

```
mass [kg] 2.82076017613
inertia [kg m^2]
[ 0.111844700704 -0.0541269993987 0.0303439737119]
[-0.0541269993987 0.26694365038 -0.0116794579938]
[ 0.0303439737119 -0.0116794579938 0.36067987241]
principal inertia [kg m^2] [0.0923910741415, 0.279146076355, 0.367931072997]
CoG offset [m] [0.0792673860649, 0.0355342175168, 0.00676203430798]
rotor force effectiveness [0.806489770793, 1.07420008532, 1.94080537622, 0.594749338359]
command delay [s] 0.02
thrust time constant [s] 0.01 (fixed)
gimbal time constant [s] 0.02 (fixed)
```

Thirteen optimized smooth coordinates

```
log_mass_scale 0.181911491571
log_second_moment_cholesky_xx_scale 0.69314718056
log_second_moment_cholesky_yy_scale 0.174301301016
log_second_moment_cholesky_zz_scale 0.69314718056
normalized_second_moment_cholesky_yx_offset 0.419680657798
normalized_second_moment_cholesky_zx_offset -0.8
normalized_second_moment_cholesky_zy_offset 0.799999999999
cog_offset_x_m 0.0792673860649
cog_offset_y_m 0.0355342175168
cog_offset_z_m 0.00676203430798
force_effectiveness_contrast_1 -0.202685330201
force_effectiveness_contrast_2 -0.599999999997
force_effectiveness_contrast_3 0.599999999998
```

Fit diagnostics

```
stitched SE(3) loss 139.244651488
full-rollout SE(3) loss 139.340706331
position RMSE [m] 0.788356622167
orientation RMSE [deg] 28.4411911201
velocity RMSE [m/s] 1.67803091031
angular velocity RMSE 1.0610800365 rad/s
specific force RMSE 1.49740067933 m/s^2
continuity max (normalized) 0.000444944568244
continuity tolerance 0.001
continuity converged True
```